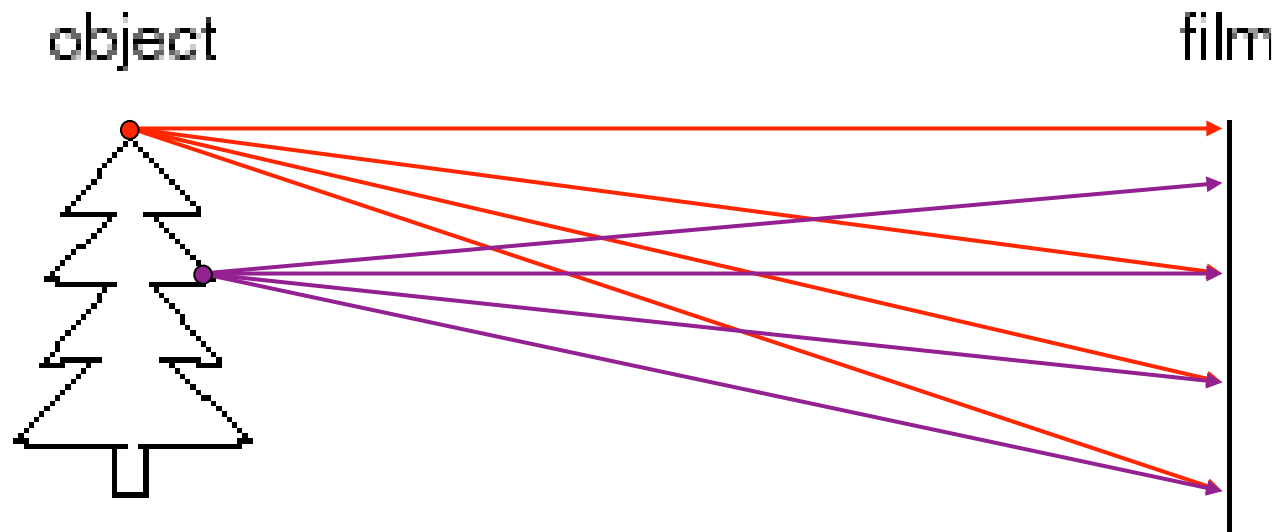


Sensors and Image Formation

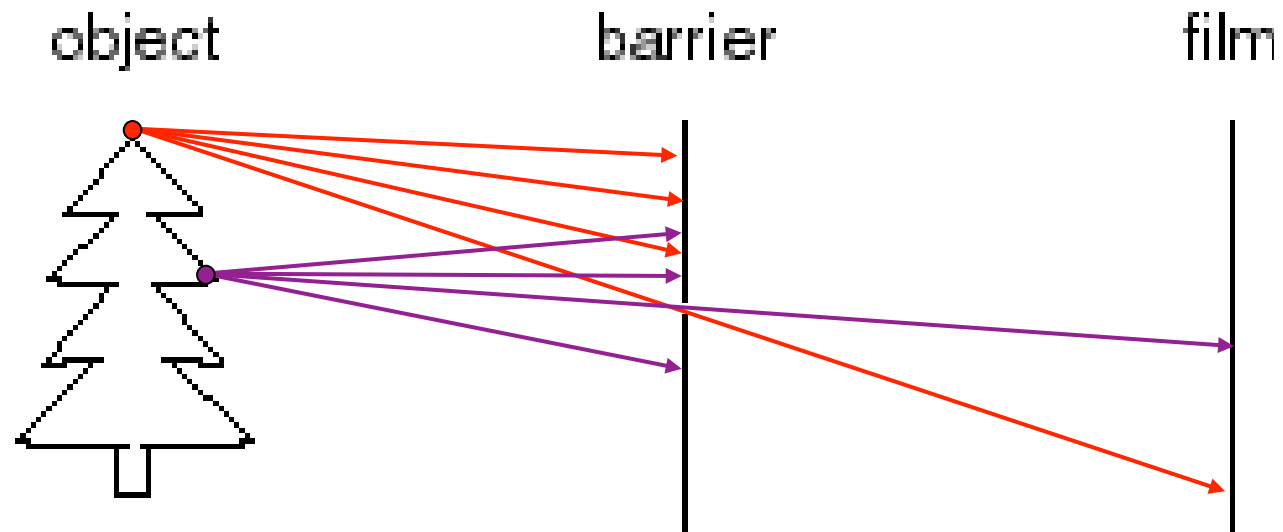
- Imaging sensors and models of image formation
- Coordinate systems

Let's design a camera



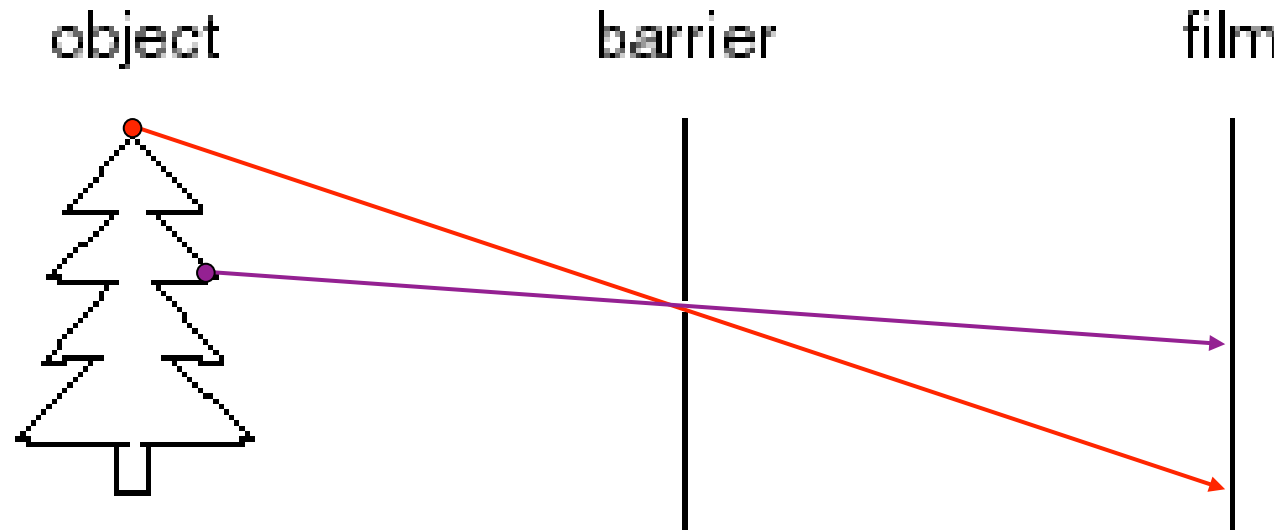
Idea 1: put a piece of film in front of an object
Do we get a reasonable image?

Pinhole camera



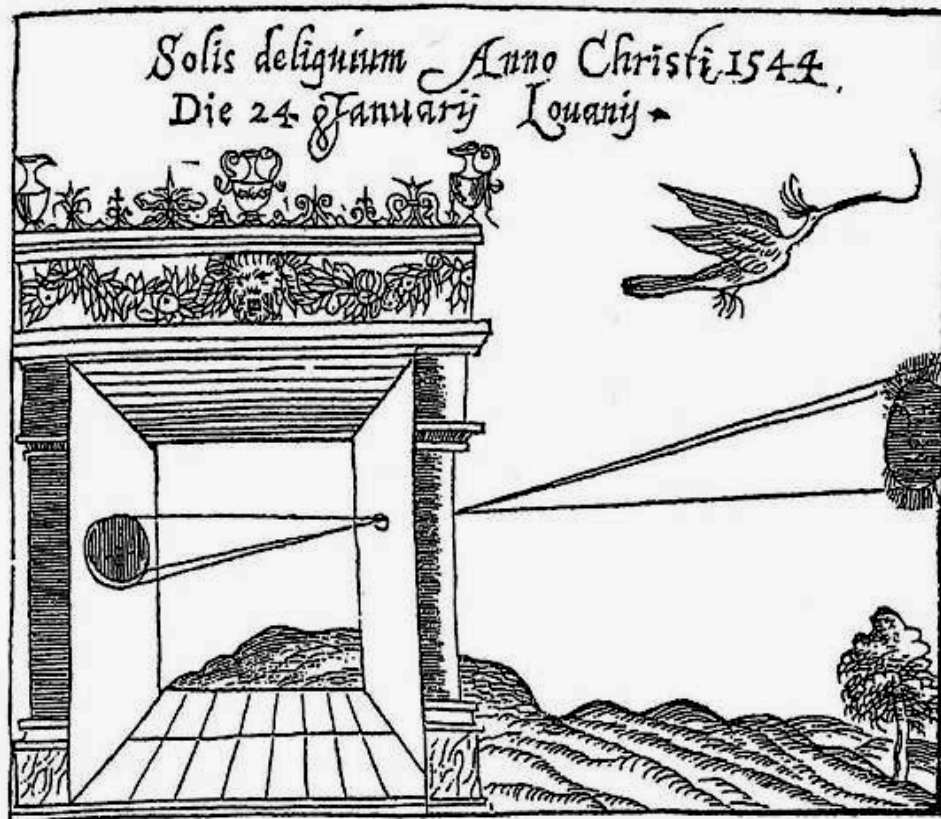
Add a barrier to block off most of the rays

Pinhole camera



- Captures **pencil of rays** – all rays through a single point: **aperture, center of projection, optical center, focal point, camera center**
- The image is formed on the **image plane**

Camera obscura



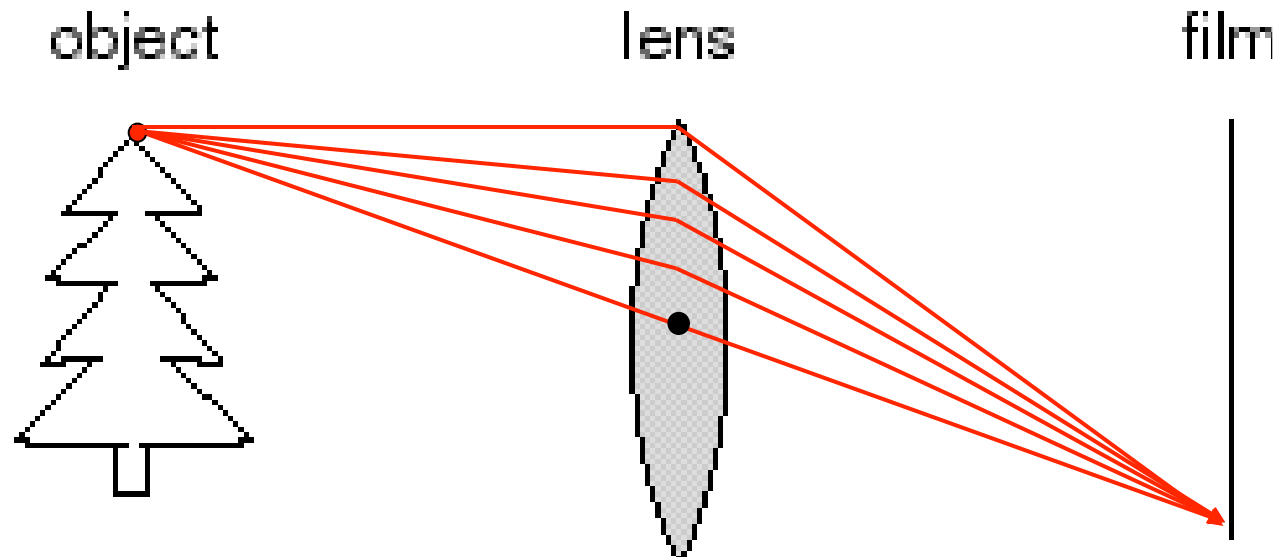
Gemma Frisius, 1558

- Basic principle known to Mozi (470-390 BCE), Aristotle (384-322 BCE)
- Drawing aid for artists: described by Leonardo da Vinci (1452-1519)

Pinhole cameras everywhere



Adding a lens

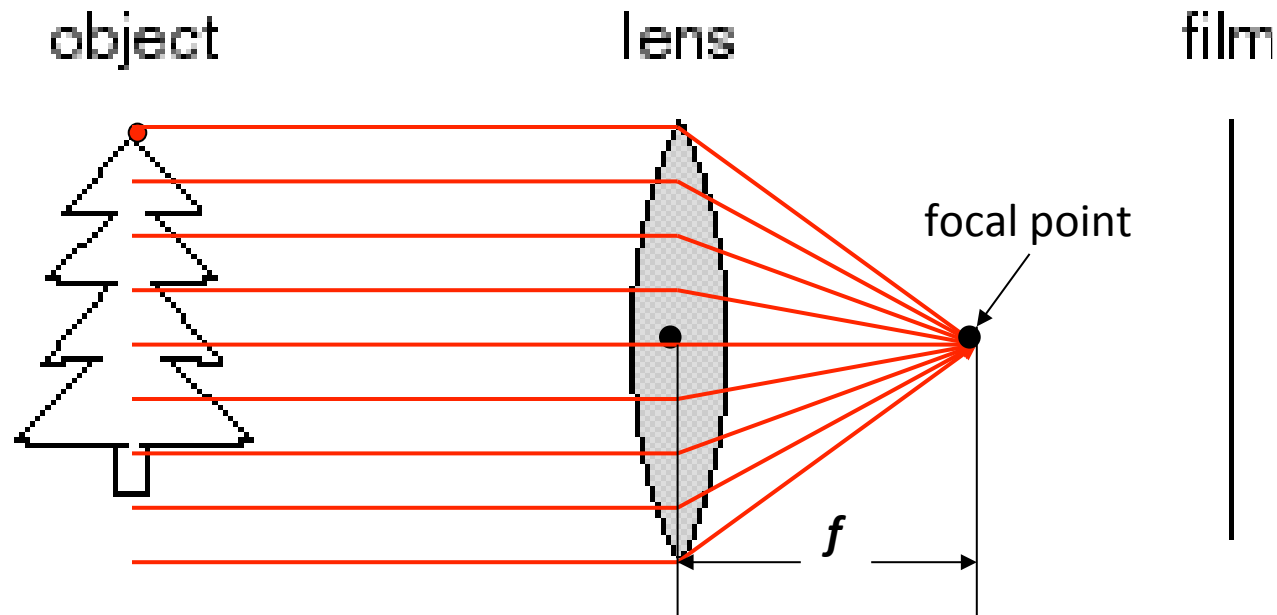


A lens focuses light onto the film

Thin lens model:

Rays passing through the center are not deviated
(pinhole projection model still holds)

Adding a lens



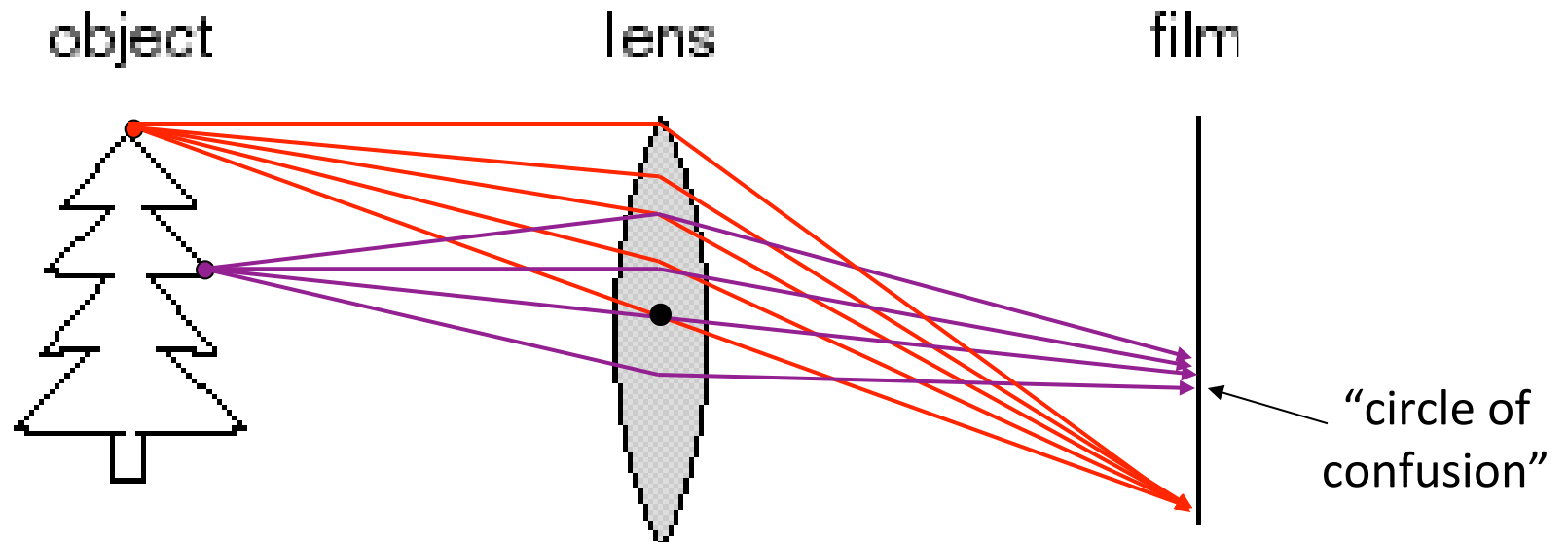
A lens focuses light onto the film

Thin lens model:

Rays passing through the center are not deviated
(pinhole projection model still holds)

All parallel rays converge to one point on a plane located at the *focal length* f

Adding a lens

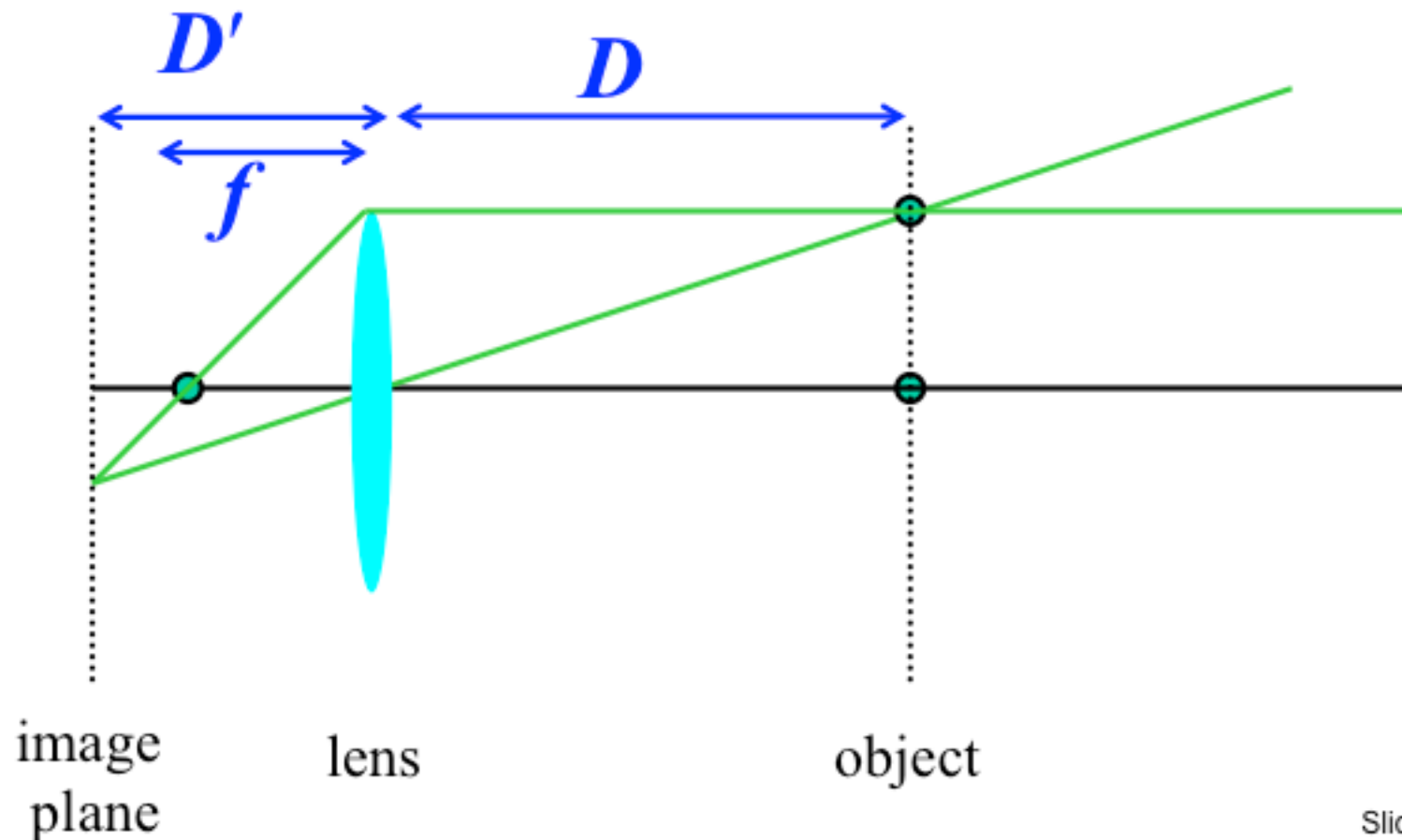


A lens focuses light onto the film

There is a specific distance at which objects are "in focus"
other points project to a "circle of confusion" in the image

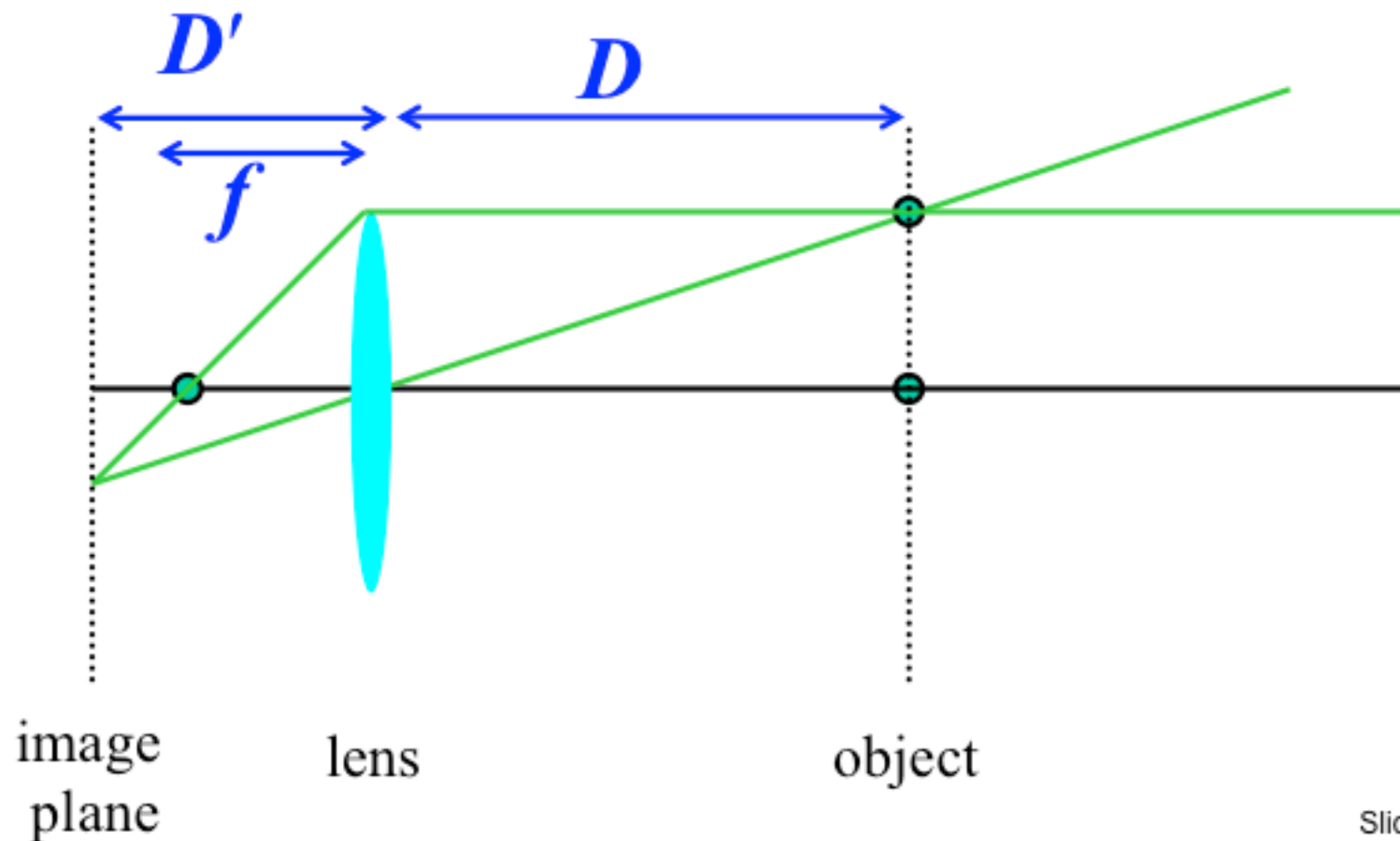
Thin lens formula

- What is the relation between the focal length (f), the distance of the object from the optical center (D), and the distance at which the object will be in focus (D')?



Thin lens formula

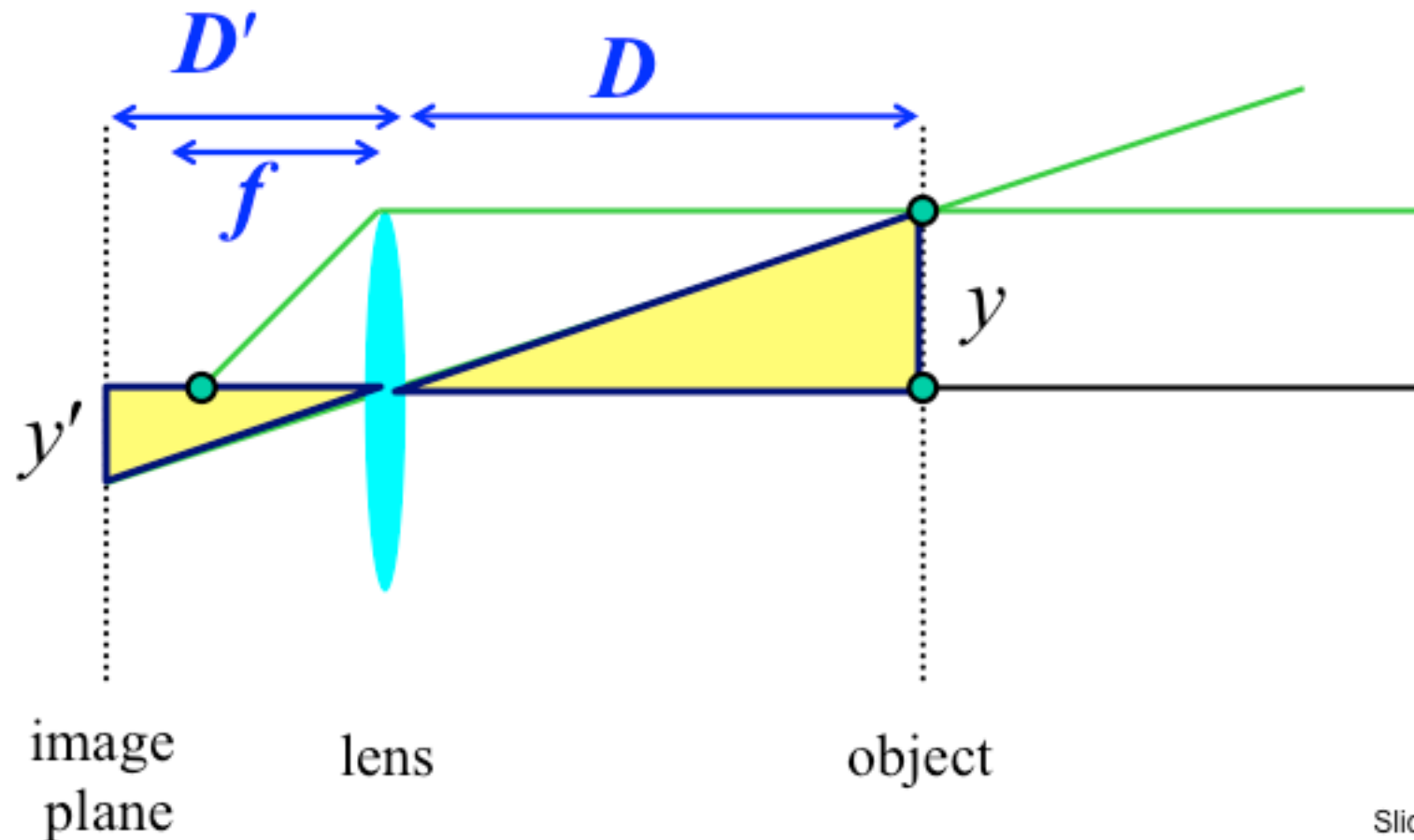
Similar triangles everywhere!



Thin lens formula

Similar triangles everywhere!

$$y'/y = D'/D$$

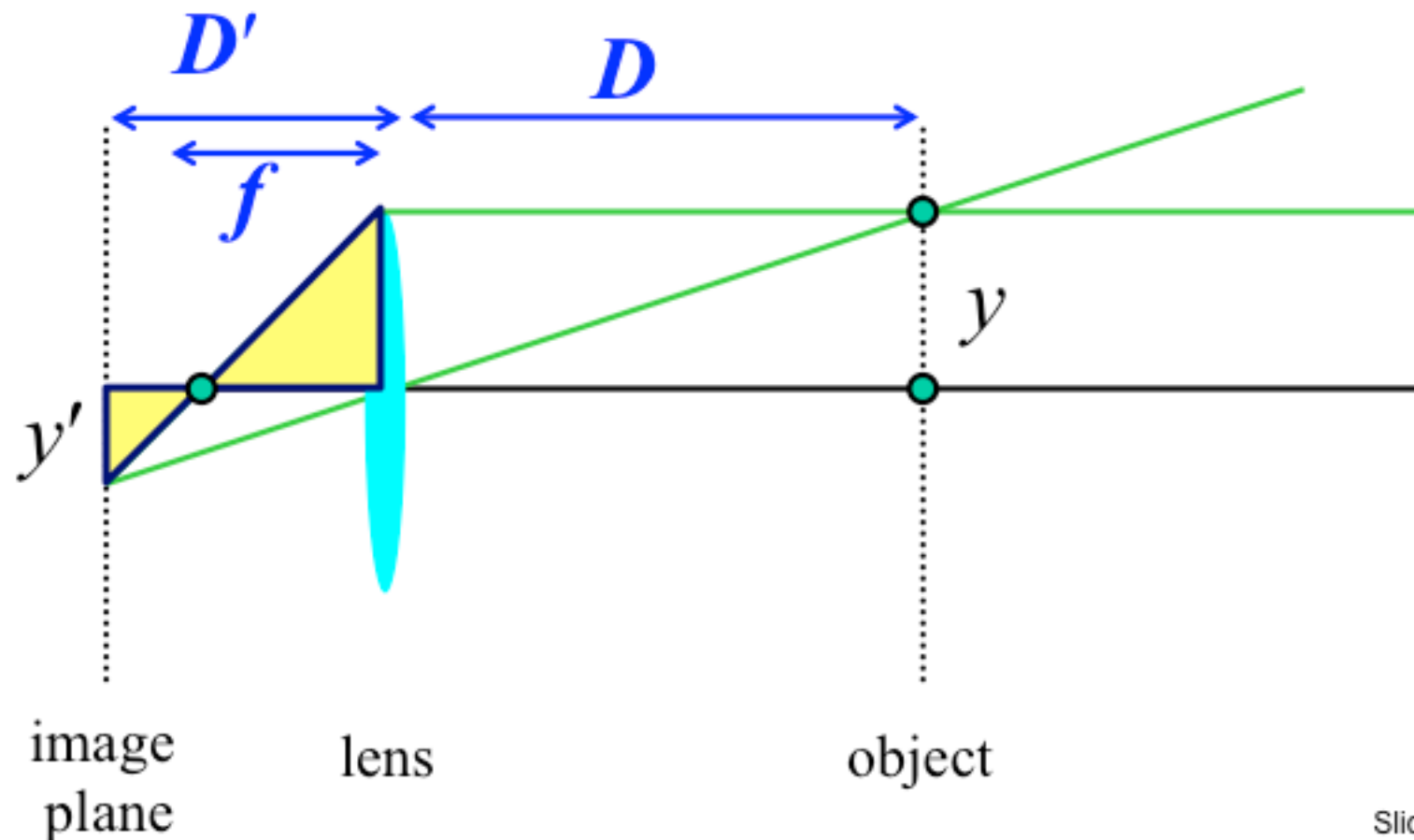


Thin lens formula

Similar triangles everywhere!

$$y'/y = D'/D$$

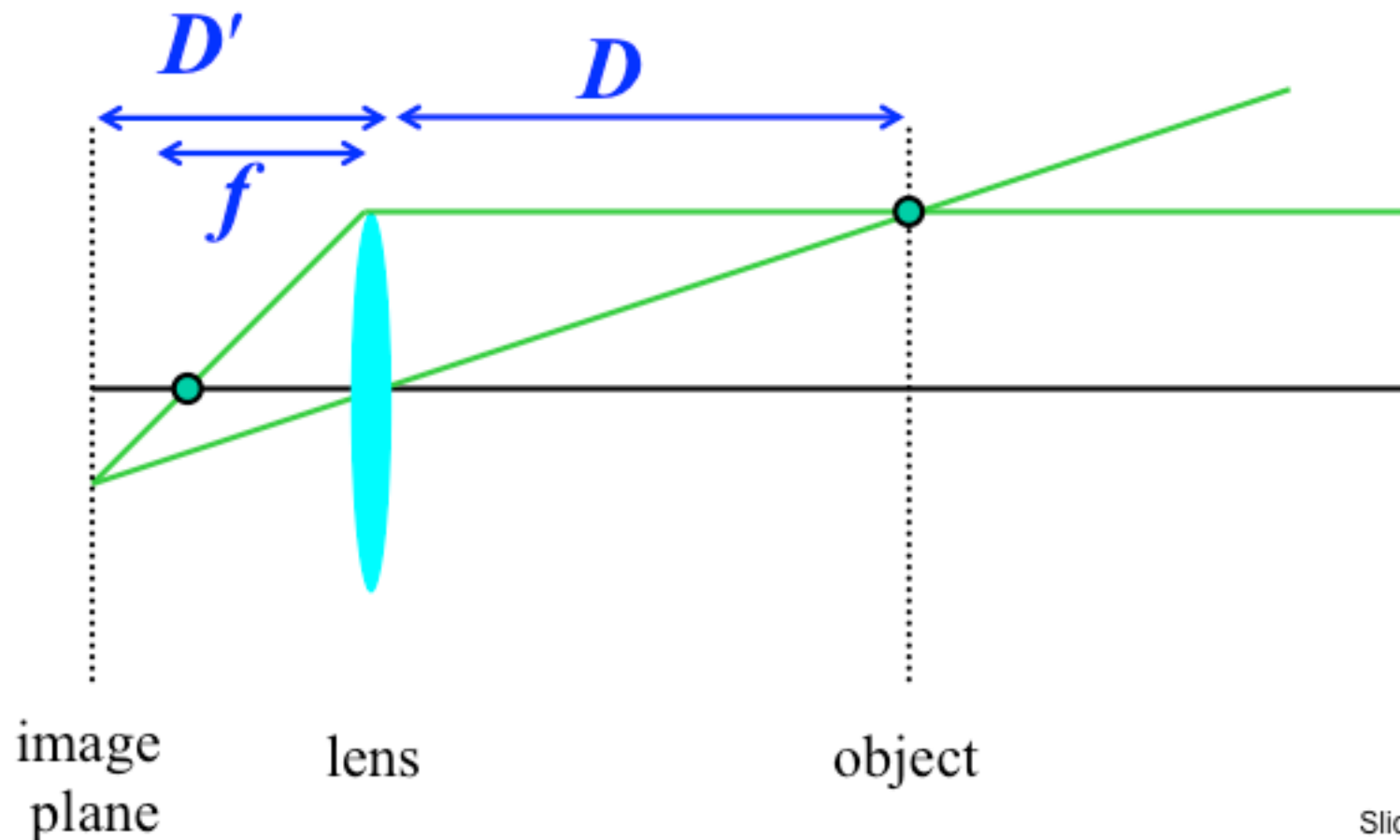
$$y'/y = (D' - f)/f$$



Thin lens formula

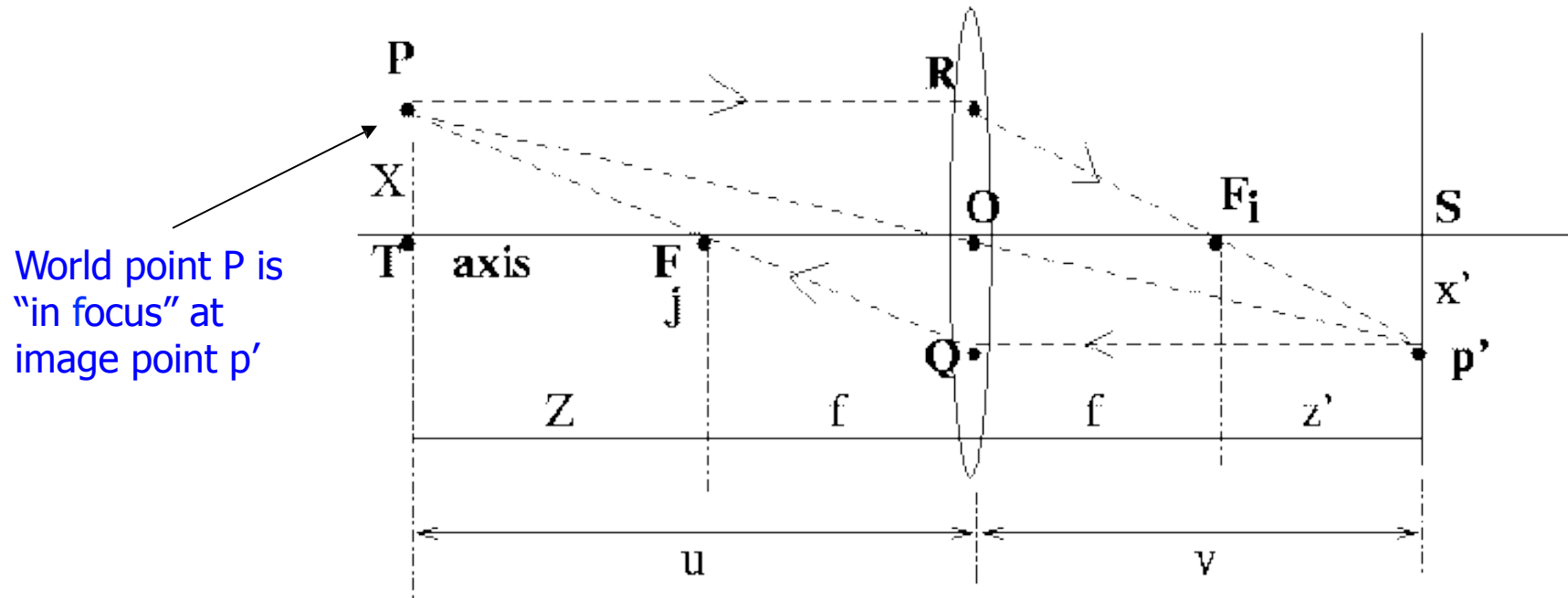
$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Any point satisfying the thin lens equation is in focus.



Thin Lens

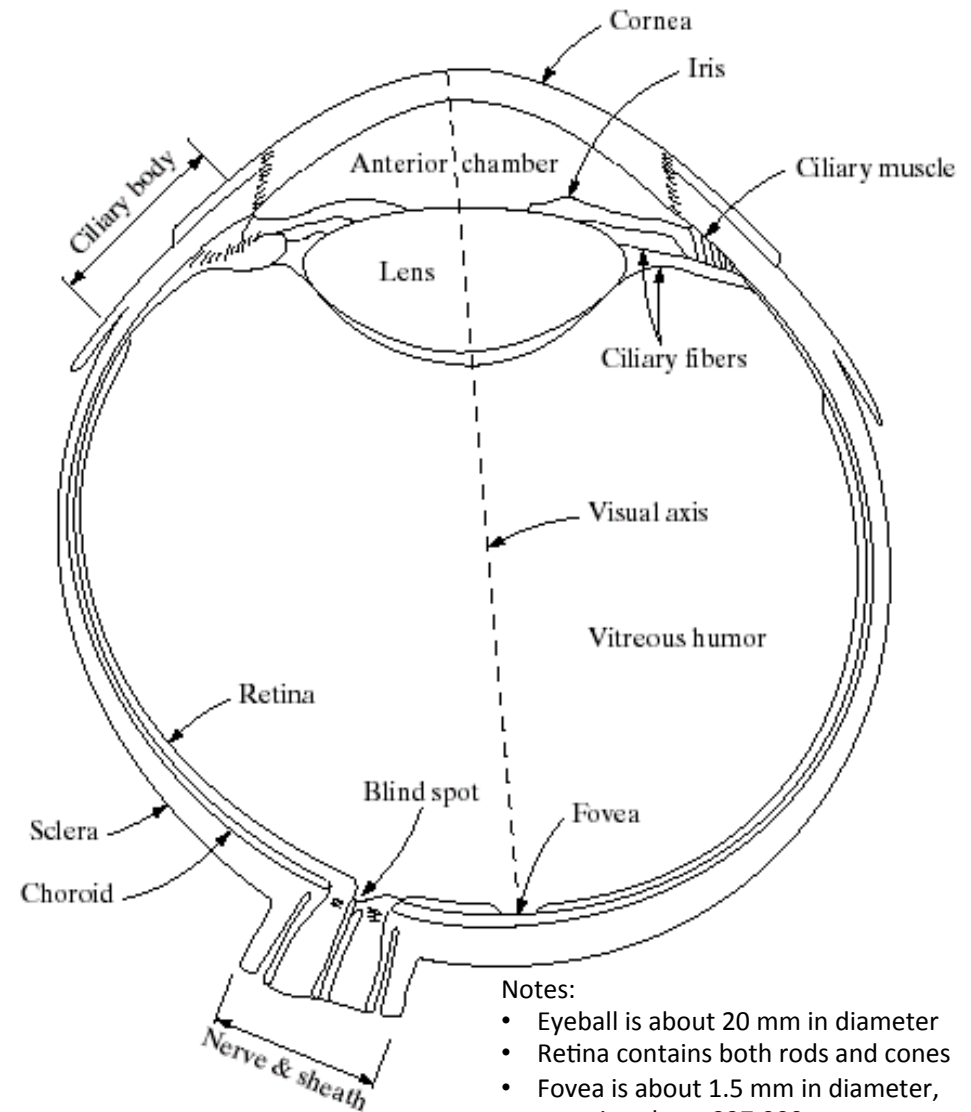
- Rays parallel to the optical axis are deflected to go through the focus
- Rays passing through the center are undeflected



Equation of a thin lens: $\frac{1}{Z} + \frac{1}{z'} = \frac{1}{f}$

Biological Vision – A Guide to Developing Computer Vision?

- Unfortunately, it is pre-attentive
 - Difficult to measure and study detailed function
- Some information from experiments with
 - Animals (electrophysiology)
 - People's perception (psychophysics)
- A great deal of processing is done right in the retina
 - Data reduction of ~100 million receptors down to ~1 million optic nerve channels
- Further processing done in the visual cortex
 - Specialized detectors for motion, shape, color, binocular disparity
 - Evidence for maps in the cortex in correspondence to the image
- Still, it is an existence proof that it can be done, and well



Digital Camera

- Image plane is a 2D array of sensor elements
- CCD type (Charge coupled device)
 - Charge accumulates during exposure
 - Charges are transferred out to shift registers, digitized and read out sequentially
- CMOS type (complementary metal oxide on silicon)
 - Light affects the conductivity (or gain) of each photodetector
 - Digitized and read out using a multiplexing scheme
- Main design factors
 - Number and size of sensor elements Chip
 - size ADC
 - resolution

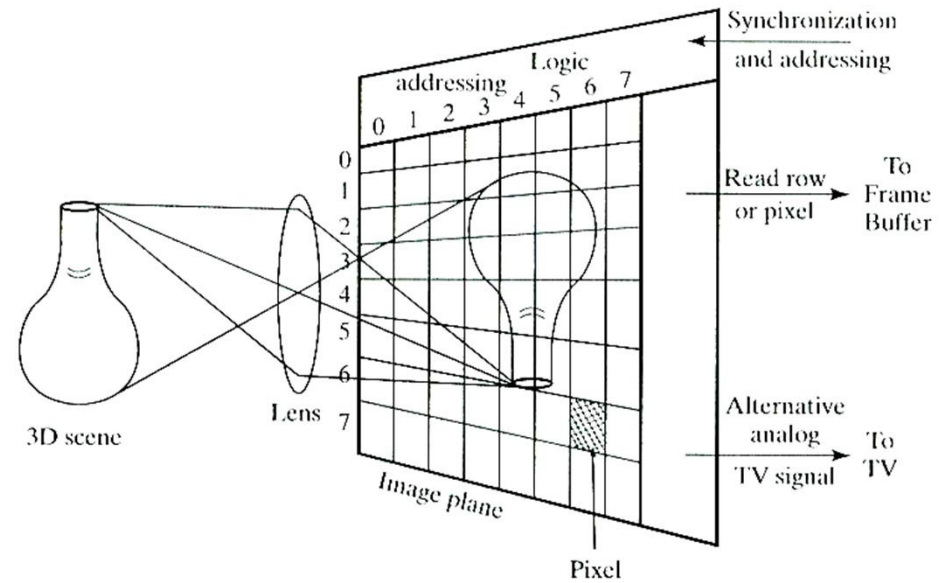


Figure 2.2 A CCD (charge-coupled device) camera imaging a vase; discrete cells convert light energy into electrical charges, which are represented as small numbers when input to a computer.

Digital Images

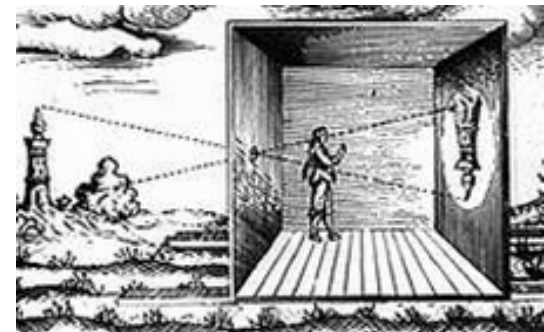
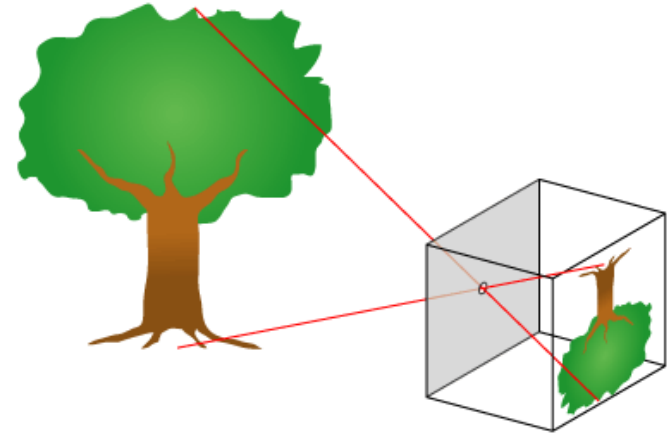
- Digital images are stored as arrays of numbers
- Numbers can represent
 - Intensity (gray level, or each color band)
 - Range
 - X-ray absorption coefficient
 - etc



10	11	10	9	9	9	10	11	12	16	18	10	9	9	9	10
10	10	10	10	10	11	10	16	26	59	69	16	10	11	9	10
10	10	10	11	16	27	49	62	89	134	147	34	12	11	15	15
10	10	11	20	43	109	153	162	165	175	171	110	22	47	73	39
9	10	37	117	166	184	187	193	180	170	171	166	65	84	65	14
10	43	165	186	185	185	189	181	158	115	135	154	123	92	16	16
35	159	183	178	174	155	118	90	77	44	28	77	138	45	51	88
79	176	186	174	150	102	78	56	35	19	14	43	102	47	146	102
89	177	186	179	175	139	104	47	25	36	90	140	141	34	135	33
98	171	181	185	189	188	158	95	68	172	198	186	188	48	84	39
114	155	177	188	192	198	193	164	154	201	209	204	210	151	43	114
142	144	167	173	178	174	172	166	178	190	202	208	209	208	115	35
150	154	161	168	168	162	176	177	175	172	183	189	203	210	171	39
155	151	162	170	164	177	186	183	167	138	173	190	193	209	175	40

Pinhole Camera Model

- A good lens can be modeled by a pinhole camera; ie., each ray from the scene passes undeflected to the image plane
- Simple equations describe projection of a scene point onto the image plane (“perspective projection”)
- We will use the pinhole camera model exclusively, except for a little later in the course where we model lens distortion in real cameras

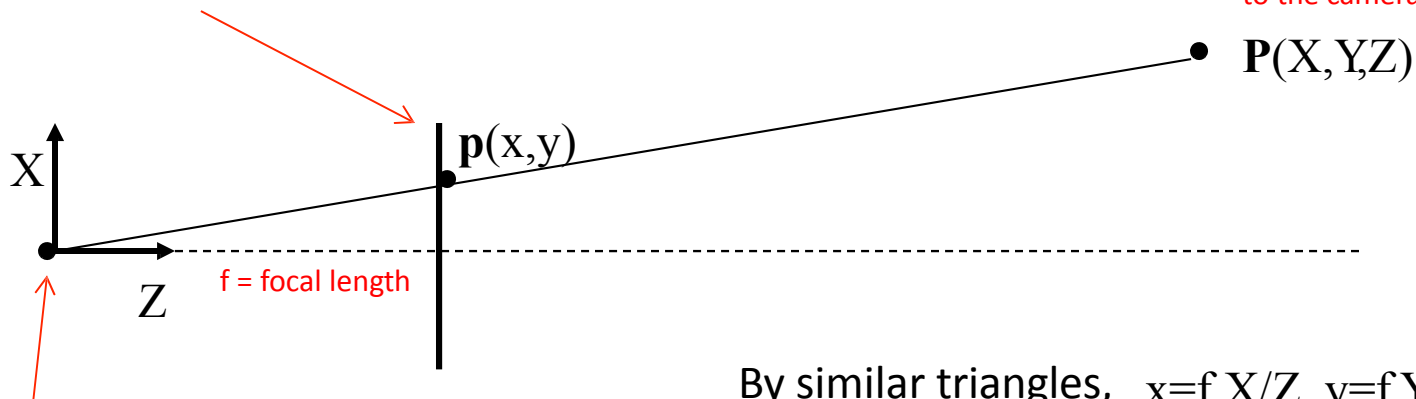


The pinhole camera (“camera obscura”) was used by Renaissance painters to help them understand perspective projection

Perspective Projection Equations

For convenience (to avoid an inverted image) we treat the image plane as if it were in front of the pinhole

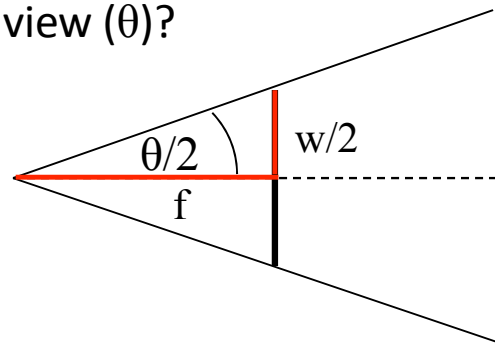
The XYZ coordinates of the point are with respect to the camera origin



We define the origin of the camera's coordinate system at the pinhole (note – this is a 3D XYZ coordinate frame)

By similar triangles, $x=f X/Z$, $y=f Y/Z$

Field of view (θ)?



$$\tan(\theta/2) = (w/2)/f$$

Camera vs Image Plane Coords

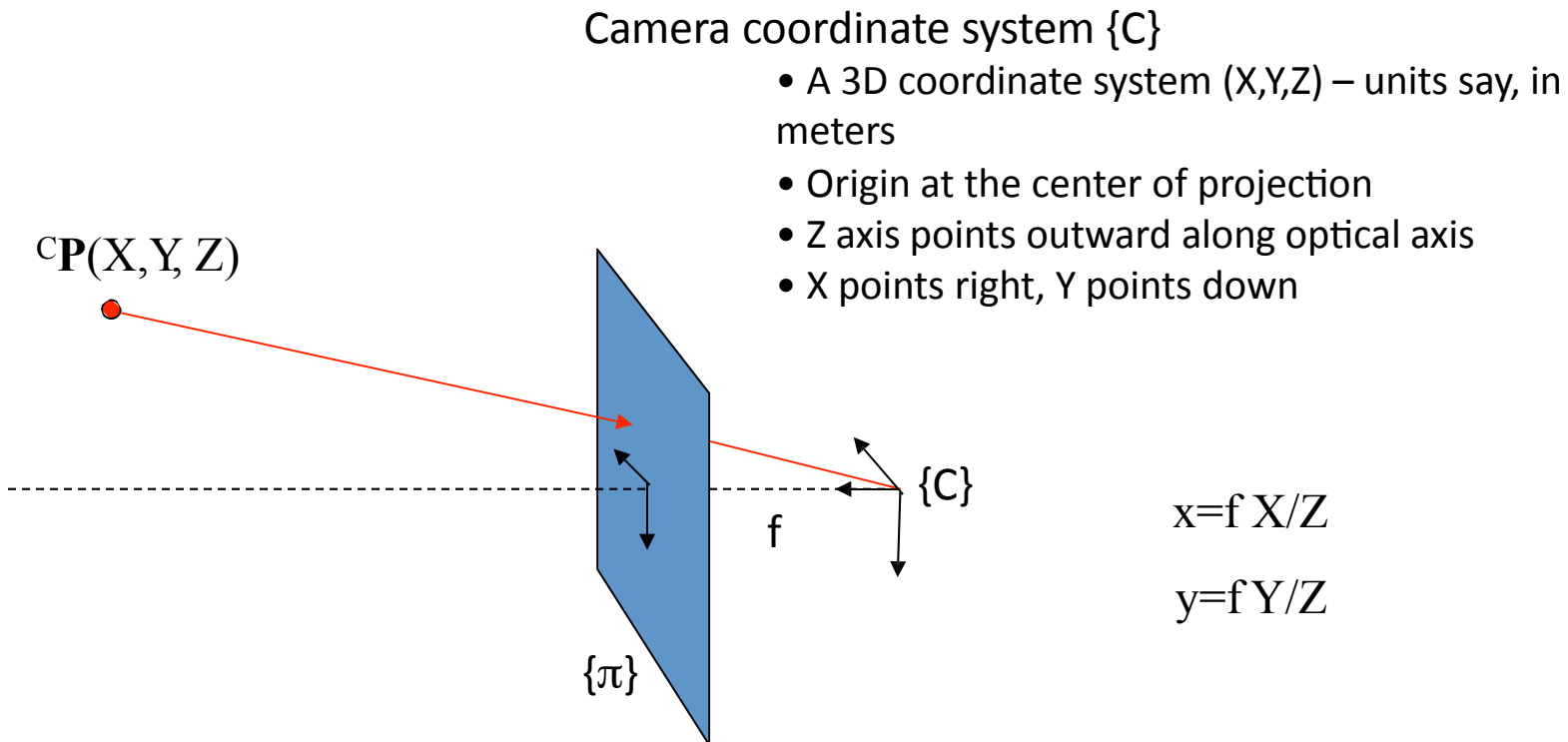


Image plane coordinate system $\{\pi\}$

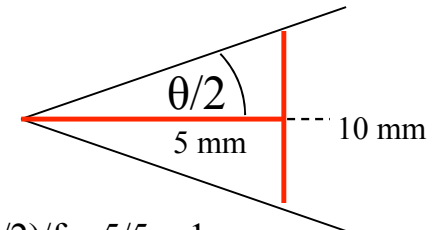
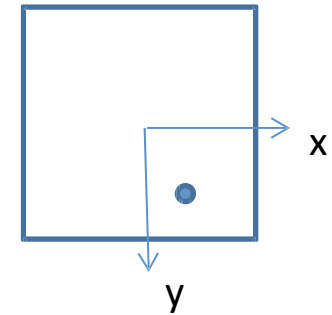
- A 2D coordinate system (x,y) – units in mm
- Origin at the intersection of the optical axis with the image plane
- In real systems, this is where the CCD or CMOS plane is

Examples

- Assume focal length = 5 mm
- A scene point is located at $(X,Y,Z) = (1\text{m}, 2\text{m}, 5\text{m})$
 - What are the image plane coordinates (x,y) in mm?
 - If the image plane is 10mm x 10mm, what is the field of view?
 - A building is 100m wide. How far away do we have to be in order that it fills the field of view?

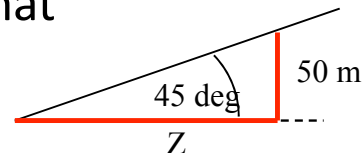
$$x = f X/Z = (5 \text{ mm}) (1 \text{ m})/(5 \text{ m}) = 1 \text{ mm}$$

$$y = f Y/Z = (5 \text{ mm}) (2 \text{ m})/(5 \text{ m}) = 2 \text{ mm}$$



$$\tan(\theta/2) = (w/2)/f = 5/5 = 1$$

$$\text{so } \theta/2 = 45 \text{ deg, fov is } 90 \times 90 \text{ deg}$$

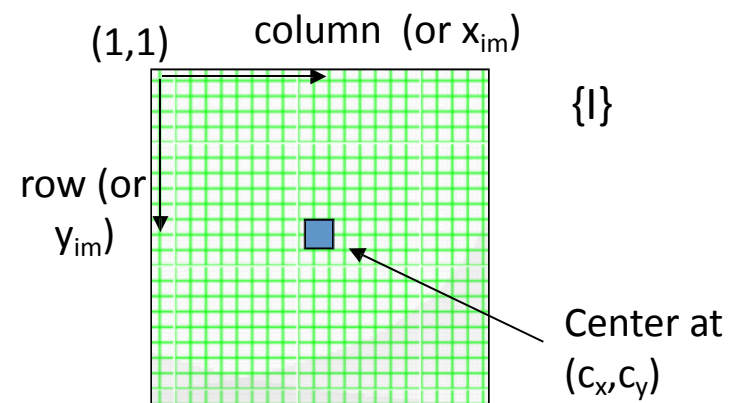
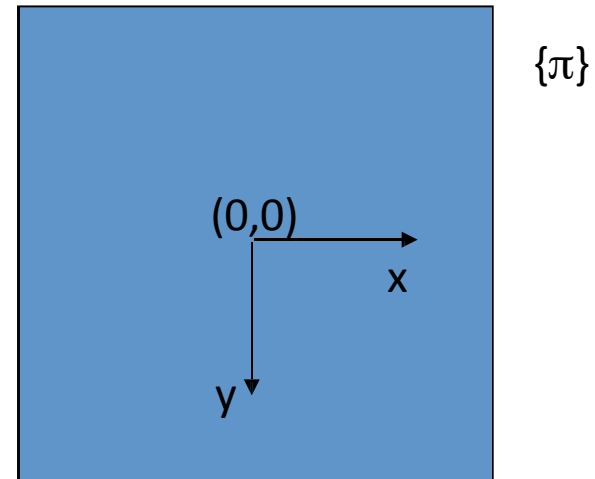


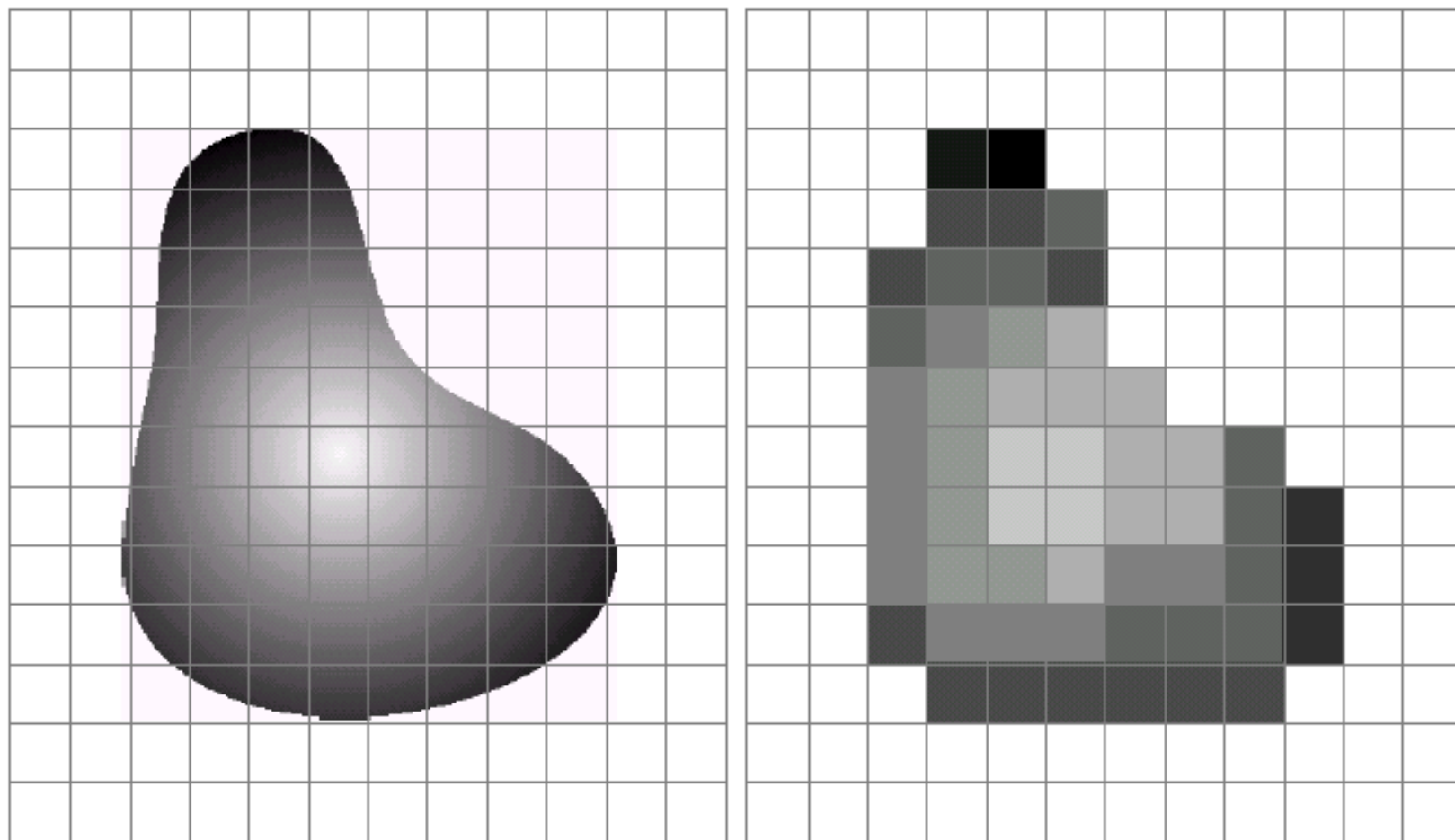
$$\tan(45 \text{ deg}) = (W/2)/Z = 50/Z$$

$$\text{so } Z = 50 \text{ m}$$

Image Buffer

- Image plane
 - The real image is formed on the CCD plane
 - (x,y) units in mm
 - Origin in center (principal point)
- Image buffer
 - Digital (or pixel) image
 - (row, col) indices
 - We can also use (x_{im}, y_{im})
 - Origin in upper left



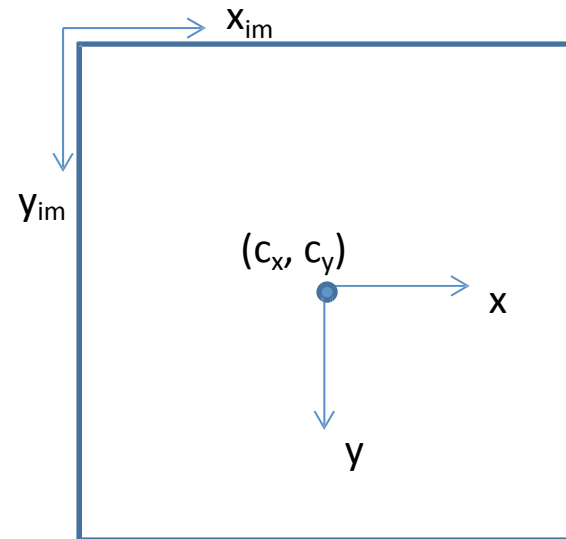


a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

Conversion between real image and pixel image coordinates

- Assume
 - The image center (principal point) is located at pixel (c_x, c_y) in the pixel image
 - The spacing of the pixels is (s_x, s_y) in millimeters



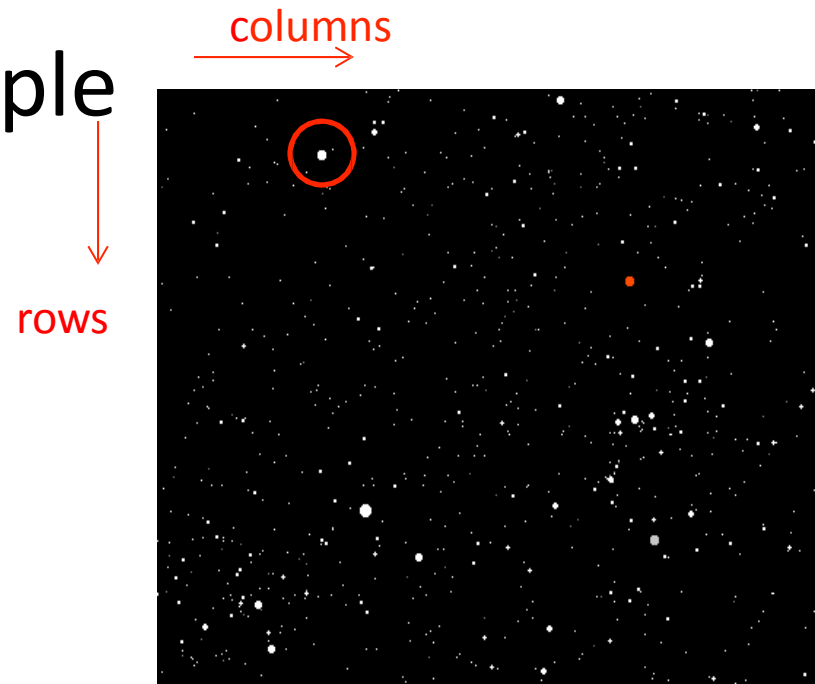
- Then

$$x = (x_{im} - c_x) s_x \quad x_{im} = x/s_x + c_x$$

$$y = (y_{im} - c_y) s_y \quad y_{im} = y/s_y + c_y$$

Example

- A star is located at pixel $(r,c)=(100,200)$ in a telescope image
- What is the 3D unit vector pointing at the star?
- Assume:
 - Image is 1000x1000 pixels
 - Optical center is in the center of the pixel image
 - CCD plane is 10 mm x 10 mm
 - Focal length is 1 m



Note on focal length

- Recall

$$x = (x_{im} - c_x) s_x$$

$$y = (y_{im} - c_y) s_y$$

- or

$$x_{im} = x/s_x + c_x$$

$$y_{im} = y/s_y + c_y$$

- and

$$x = f X/Z$$

$$y = f Y/Z$$

- So

$$x_{im} = (f / s_x) X/Z + c_x$$

$$y_{im} = (f / s_y) Y/Z + c_y$$

- All we really need is

$$f_x = (f / s_x)$$

$$f_y = (f / s_y)$$

- We don't need to know the actual values of f and s_x, s_y ; just their ratios

- We can alternatively express focal length in units of pixels

Example

- A camera observes a rectangle 1m away
 - The rectangle is known to be 20cm x 10cm
 - In the image, the rectangle measures 200 x 90 pixels
- Focal length in x, focal length in y (pixels)?
- If image size is 640x480 pixels, what is field of view (horiz, vert)?

Camera Parameters

- Intrinsic parameters
 - Those parameters needed to relate an image point (in pixels) to a direction in the camera frame
 - f_x, f_y, c_x, c_y
 - Also lens distortion parameters (will discuss later)
- Extrinsic parameters
 - Define the position and orientation (pose) of the camera in the world

Frames of Reference

- Image frames are 2D; others are 3D
- The “pose” (position and orientation) of a 3D rigid body has 6 degrees of freedom

